

BRATSK, Russian Federation

WMO: 303090

Lat: **56.371N**Lon: **101.698E**Elev: **1610**StdP: **13.86**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.6	-29.3	-40.6	0.6	-35.1	-34.6	0.8	-29.1	21.8	5.9	18.6	4.4	3.5	290	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	82.7	65.1	79.6	64.1	76.8	62.8	68.1	78.2	66.2	76.0	64.5	73.9	6.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	96.3	73.1	62.6	90.2	71.1	60.8	84.5	69.1	33.4	78.6	31.9	76.1	30.5	73.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.1	15.2	-36.4	88.3	9.0	2.0	-42.9	89.8	-48.1	90.9	-53.2	92.1	-59.7	93.5		
(5)			-36.5	71.4	8.9	2.0	-42.9	72.8	-48.2	74.0	-53.2	75.1	-59.6	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.2	-4.2	3.0	18.0	33.5	46.9	60.5	65.3	59.9	46.1	31.4	12.4	-0.1		
	DBStd	26.20	14.36	12.87	11.53	8.80	9.02	7.82	5.72	6.81	7.99	9.68	14.20	14.04		
	HDD50	8108	1681	1317	993	499	171	11	0	9	168	577	1129	1553		
	HDD65	12485	2146	1737	1458	946	564	176	69	184	567	1041	1579	2018		
	CDD50	1243	0	0	0	2	75	324	474	315	52	1	0	0		
	CDD65	147	0	0	0	0	3	40	77	26	1	0	0	0		
	CDH74	1237	0	0	0	0	43	405	568	214	7	0	0	0		
	CDH80	243	0	0	0	0	5	84	121	33	0	0	0	0		
	WSAvg	6.6	5.5	5.9	7.0	8.3	8.1	6.2	5.6	5.7	6.8	7.3	7.1	5.8		
	PrecAvg	14.5	0.7	0.5	0.5	0.7	1.3	2.0	2.2	2.4	1.5	1.0	1.0	0.8		
(15) Precipitation	PrecMax	19.1	1.5	1.0	1.0	1.7	2.8	5.7	6.2	4.1	3.5	1.5	1.5	1.5		
	PrecMin	9.7	0.3	0.2	0.2	0.1	0.3	0.1	0.4	0.4	0.1	0.4	0.4	0.4		
	PrecStd	2.2	0.3	0.2	0.2	0.4	0.6	1.1	1.4	0.9	0.7	0.4	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	31.9	45.7	61.4	78.9	86.1	87.7	84.0	73.7	57.2	44.2	30.8		
		MCWB	26.6	27.3	36.4	45.9	58.9	64.7	68.9	66.6	58.7	46.9	39.0	27.7		
	2%	DB	23.2	27.8	41.0	56.0	73.3	82.4	83.5	79.1	67.9	52.1	37.8	25.2		
		MCWB	21.9	24.8	33.1	43.2	54.9	63.1	66.8	64.8	56.6	44.5	34.0	23.8		
	5%	DB	19.1	24.6	37.2	51.6	68.0	78.9	80.3	75.7	62.8	48.2	34.2	21.7		
		MCWB	17.8	22.5	31.1	40.9	51.7	62.3	65.7	63.2	53.5	42.2	31.3	20.4		
	10%	DB	14.1	20.8	33.6	46.7	62.8	75.2	76.9	72.1	58.8	44.4	31.3	18.5		
		MCWB	13.4	18.8	28.6	38.3	49.2	60.6	64.3	61.5	51.5	39.9	29.2	17.3		
	0.4%	WB	26.2	28.1	36.8	48.3	60.0	68.9	71.7	68.4	61.6	48.7	39.1	28.4		
		MCDB	27.4	30.6	43.8	57.3	75.6	79.9	82.6	78.8	69.5	54.4	42.4	30.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.1	25.1	33.8	44.7	56.4	66.1	69.0	66.3	57.6	45.6	34.4	24.1		
		MCDB	23.3	27.2	39.7	53.7	70.7	77.6	79.1	76.7	65.6	50.4	37.1	25.3		
	5%	WB	17.4	22.2	31.2	41.3	53.2	63.9	67.3	64.4	54.8	42.9	31.6	20.6		
		MCDB	18.4	24.4	36.5	50.1	64.7	75.3	76.8	73.9	61.9	47.6	33.9	21.7		
	10%	WB	13.2	18.4	28.7	38.5	50.3	61.9	65.6	62.5	52.3	39.8	28.9	17.3		
		MCDB	13.9	20.4	33.3	46.2	61.4	72.2	74.4	70.8	57.8	43.5	31.2	18.4		
	5% DB	MDBR	12.3	15.3	17.7	16.6	19.3	19.5	18.0	17.6	15.9	12.9	12.8	12.1		
		MCDBR	13.5	15.2	19.1	22.0	26.2	24.8	22.3	22.6	21.7	18.4	11.7	12.9		
		MCWBR	12.9	13.2	13.8	13.7	13.7	11.6	10.9	11.5	12.6	12.1	10.0	12.1		
		MCWBR	13.0	15.3	17.7	20.2	24.1	21.5	20.0	20.8	20.5	16.5	11.9	13.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	12.5	13.6	13.4	13.5	13.8	11.3	10.9	11.3	12.5	11.5	10.7	12.5		
	Clear-Sky Solar Irradiance	taub	0.272	0.300	0.326	0.359	0.395	0.416	0.434	0.392	0.342	0.329	0.287	0.264		
		taud	2.018	2.029	2.053	2.216	2.237	2.257	2.225	2.326	2.432	2.286	2.090	1.977		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	186	232	259	271	267	262	254	259	258	224	179	152		
		Edh at Noon	20	30	38	38	40	40	41	34	27	24	19	16		
(44) All-Sky Solar Radiation	RadStd		250	562	1023	1446	1651	1834	1687	1373	886	469	239	156		
			24	33	59	99	148	134	145	87	104	49	27	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page